AI-Driven Clinical Intelligence Framework for Early-Stage Pulmonary Malignancy Detection and Therapeutic Guidance

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Abstract—Pulmonary malignancies remain a serious issue in the domain of oncology across the world due to the time lag in diagnosis and absence of screening. This paper presents an intelligent clinical decision support system using deep learning neural networks for the analysis of computerized tomography scans. The proposed system uses convolutional deep learning for the analysis of radiological images, thus improving the accuracy of cancer pattern identification. The system offers a multi-level risk stratification and offers in-depth clinical information, including treatment, prevention, and donor evaluation processes. The interpretability of the system is improved using explainable AI (XAI) techniques. Experimental outcomes confirm the accuracy of 95.2% and AUC-ROC of 96.1%, thus proving the system's superior performance.

Index Terms—pulmonary malignancy, convolutional neural networks, CT scan analysis, clinical decision support system, explainable AI, risk stratification, Grad- CAM

I. Introduction

Pulmonary neoplasms are one of the most difficult health issues in the world, and they also contribute significantly to the mortality rate of patients with oncological diseases. The main reason for the high mortality rate is the presentation of advanced stages of the disease. The traditional methods used in the diagnosis of pulmonary neoplasms are time-consuming, expensive, and require specialized skills..

A. Clinical Context and Significance The rapidly evolving field of Artificial Intelligence has triggered the use of machine learning and deep learning approaches in medical science to enhance diagnostic precision and efficiency. Specifically, the use of convolutional neural networks has demonstrated a strong capability in the automated analysis of medical images to accelerate and enhance the accuracy of radiological abnormalities in computerized tomography scans. Most of the modern AI-based lung cancer diagnosis tools are only used for the purpose of cancer diagnosis and do not offer any further clinical advice. Most of the current methods are black-box models that lack interpretability.

B. Research Motivation

In order to fill these important research gaps, we propose an integrated clinical intelligence system that incorporates automated radiological analysis and risk analysis with interpretable decision support mechanisms. Our proposed solution seeks to improve the accuracy of diagnosis, assist medical professionals, and ultimately improve patient outcomes through multi-dimensional analysis that includes image-based detection, patient risk stratification, treatment pathway recommendation, and preventive advice

II. Literature Analysis

Recent developments in artificial intelligence and computational learning algorithms have been extensively researched for the detection and diagnosis of lung cancer. Various research studies have validated the efficiency of AI algorithms in processing medical images and data. However, various issues and concerns have been raised regarding the interpretability of AI models and decision support.

Alowidi et al. [1] presented a learning algorithm for predicting cancer from patient clinical data, which gave better classification results than statistical methods. However, the proposed system did not use XAI methods or decision support systems. The application of decision support systems based on AI in the healthcare industry was investigated by Freedman et al. [2] in the context of organ transplantation workflow. Topuz et al.

[3] proposed models for prediction using EHRs but did not use CT scan analysis or offer personalized suggestions. Shen et al. [4] proposed multi-scale CNNs for lung nodule classification with high accuracy but without interpretability. Lundervold and Lundervold [5] presented a review of deep learning in medical imaging. A hybrid framework of DenseNet169+SVM with XAI was proposed by Islam et al. [8]. Zhu et al. [9] introduced DeepLung, a 3D dual-path network for automatic nodule detection.

A. Research Gaps Identified

Based on the analysis of the current literature, the following research gaps have been identified:

- The current literature is dominated by studies that assess the accuracy of detection based on either clinical data or imaging data, but not both.
- There are no comprehensive decision support systems available that combine both clinical and imaging data.
- The current literature does not have the ability to conduct risk

assessment and stratification.

- There is no existing research that assesses the functionality of donor screening.
- The use of Explainable AI in a clinical environment is

very limited.

III. Proposed Intelligent Framework

The system consists of an AI-based prediction structure for pulmonary cancers combined with clinical decision support. This approach enhances chances of early-stage diagnosis, estimates the patient's risk of developing the disease, and assists clinicians with informed decisions.

A. Automated Radiological Image Processing

The first component applies deep learning algorithms to assist doctors in diagnosing pulmonary cancers based on abnormal radiological inputs. CNNs accurately extract spatial and contextual features from the CT image input, enabling the verification and identification of cancerous areas based on objective image inputs.

Image pre-processing operations like noise removal, normalization, standardization of dimensions, and segmentation are imposed to ensure high-quality images. There is less dependence on human analysis by radiologists, and there is consistency in the results of diagnosis with reduced human error.

B. Hazard Assessment and Clinical Guidance

The intelligent framework carries out a comprehensive risk analysis based on the allocation of patient categories into four levels: minimal, moderate, elevated, or critical. Risk stratification is based on the prediction probability (confidence), derived imaging features, and the clinical parameters provided by the physician. The framework provides recommendations on the options for follow-up care, the frequency of care, and the diagnostic tests to be conducted.

C. Therapeutic Planning and Preventive Strategies

By taking into account the characteristics of the tumor, the level of risk, and the health status of the patient, the system generates a personalized treatment plan. The treatment options can be surgery, chemotherapy, radiation therapy, or molecular targeted therapy. The system also gives recommendations on how patients with high risk can avoid the development of cancer in the future. The system can also screen possible organ donors based on the health of the lungs

IV. System Architecture

The proposed intelligent system for Prediction of Pulmonary Malignancy (PM) and Intelligent Clinical Decision Support (ICDS) is composed of several interlinked modules that analyze the medical data in a sequential and efficient way. The overall system architecture is depicted in Fig. 1.

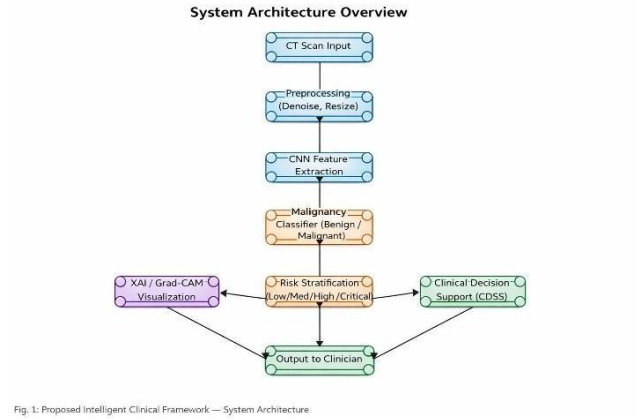

Fig. 1. System Architecture of the Proposed Intelligent Clinical Framework

A. Data Acquisition Module

The module acquires images from the CT scan available in public medical imaging repositories and hospital websites, along with clinical information such as age, smoking status, symptoms, and medical history.

B. Data Preprocessing Module

The acquired data is preprocessed to enhance quality and uniformity. For the CT scan images, this includes noise reduction, normalization, resizing, and contrast enhancement. For the clinical information, this includes missing data and feature normalization.

C. Feature Extraction Component

This component uses CNN-based deep learning techniques to extract key features from the processed images of the CT scan. The algorithm extracts features related to texture,

shape, and location that are required for the detection of pulmonary cancers.

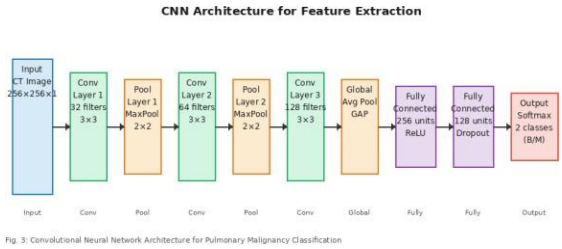

Fig. 3. CNN Architecture for Feature Extraction and Malignancy Classification

D. Prediction and Risk Assessment Component

The trained AI models on the extracted features are utilized for the classification of each input as either benign or malignant, along with a risk severity level based on the classifier confidence and CNN features.

E. Clinical Decision Support System (CDSS)

The CDSS provides clinical actions based on the hazard level, such as additional tests, follow-up, treatment, and prevention. The XAI method is used to represent and explain the recommendations of the CDSS.

F. Output and User Interface

The user interface provides the prediction output, risk level, and recommended actions in a readable manner, which is easy for the doctor to understand and act upon.

V. METHODOLOGY

The overall end-to-end flowchart of the proposed system methodology from the raw input CT scans to output clinical reports.

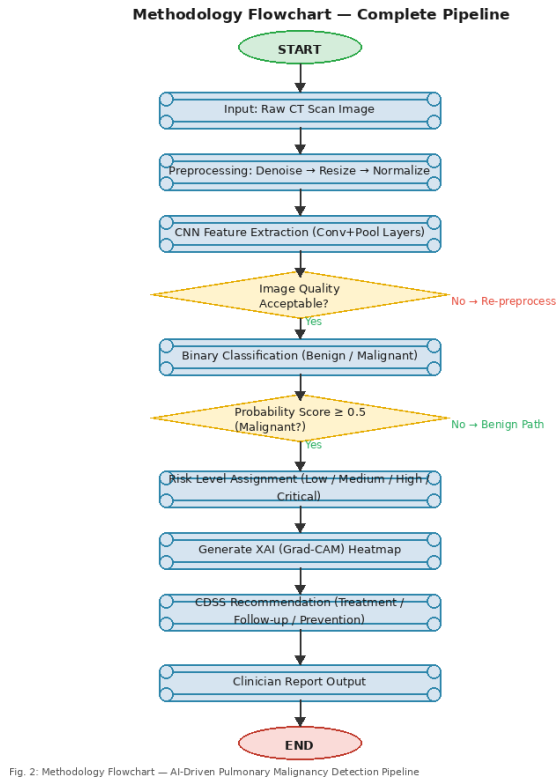

Fig. 2. Methodology Flowchart — Complete AI-Driven Detection Pipeline

A. Dataset Description

A. Dataset Description

The dataset is comprised of 1,000 pulmonary CT scan images sourced from publicly available medical datasets and clinical repositories, with 500 images of malignant and 500 images of benign conditions. The dataset is split into 80% for training and 20% for testing, allowing assessment on unseen data.Image Preprocessing

Each of the CT scan images is subjected to a preprocessing step prior to feature extraction. The transformation can be mathematically modeled as:

$$I_processed = Normalize(Resize(Denoise(I_raw))) \quad (1)$$

where I_raw is the original CT scan image and I_processed is the preprocessed image. The denoising step involves the application of a Gaussian smoothing filter:

$$I_denoised(x,y) = \int I_raw(x',y') \cdot G\sigma(x-x', y-y')\, dx'dy' \quad (2)$$

where Gσ is the Gaussian kernel with standard deviation σ. The normalization step scales the pixel values to the range [0,1]:

$$I_norm(x,y) = (I(x,y) - I_min) / (I_max - I_min) \quad (3)$$

B. CNN-Based Feature Extraction

The CNN uses successive layers of convolution, activation, and max-pooling to detect both low-level features (edges, contours) and high-level semantic features (nodule shape, density patterns). The convolution process at layer l is given by:

$$F_l(x,y) = \sigma(\sum_{i,j} W_l(i,j) \cdot F_{l-1}(x+i, y+j) + b_l) \quad (4)$$

where W_l and b_l are the weights and biases to be learned at layer l, and σ is the ReLU activation function, which is defined as:

$$\sigma(z) = \max(0, z) \quad (5)$$

The max-pooling process shrinks the spatial dimension while retaining the prominent features:

$$P_l(x,y) = \max_{(i,j)\in R(x,y)} F_l(i,j) \quad (6)$$

where R(x,y) is the region

C. Pulmonary Malignancy Classification

After feature extraction, the CNN applies binary classification with a softmax output layer. The probability of malignancy P(M|x) given input feature vector x is given by:

$$P(M|x) = \exp(z_M) / (\exp(z_M) + \exp(z_B)) \quad (7)$$

where z_M and z_B are the output logits for malignant and benign classes respectively. The model is trained with categorical cross-entropy loss:

$$L_{CE} = -(1/N) \sum_{i=1}^{N} [y_i \log(\hat{y}_i) + (1-y_i) \log(1-\hat{y}_i)] \quad (8)$$

where N is the number of training samples, y_i is the ground truth label, and $\hat{y}_i$ is the predicted probability. The Adam optimizer dynamically adjusts the learning rate:

$$m_t = \beta_1 m_{t-1} + (1-\beta_1) \nabla L \quad (9)$$

$$v_t = \beta_2 v_{t-1} + (1-\beta_2) (\nabla L)^2 \quad (10)$$

$$\theta_{t+1} = \theta_t - \alpha \cdot \hat{m}_t / (\sqrt{\hat{v}_t} + \varepsilon) \quad (11)$$

where β_1=0.9, β_2=0.999 are moment decay rates, α is the learning rate, and $\varepsilon=10^{-8}$

D. Risk Assessment Mechanism

Following the classification process, a risk assessment module is employed to categorize patients into four risk levels using the classifier's probability output P(M|x). Fig. 4 depicts the risk level stratification decision tree. The overall risk score R is calculated as:

$$R = \alpha_1 P(M|x) + \alpha_2 C_{age} + \alpha_3 C_{smoke} + \alpha_4 C_{hist} \quad (12)$$

where P(M|x) is the malignancy probability obtained using the CNN, C_{age} is the age risk factor coefficient normalized to the range [0, 1], C_{smoke} is the smoking history coefficient, C_{hist} is the clinical history factor, and α_1-α_4 are empirically set weight coefficients summing to 1. The risk levels are determined as follows:

Tier = Low, if $R < 0.35$

Tier = Medium, if $0.35 \le R < 0.55$

Tier = High, if $0.55 \le R < 0.75$

Tier = Critical, if $R \ge 0.75$ (13)

The risk level stratification process helps doctors in risk assessment to prioritize patients and select those who need immediate attention.

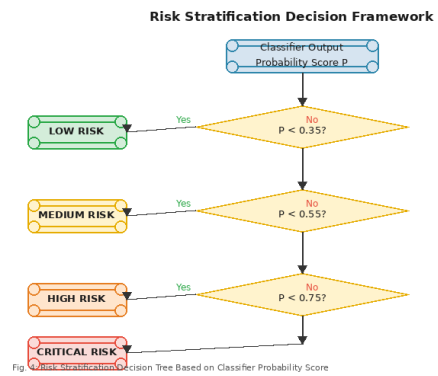

Fig. 4. Risk Stratification Decision Tree Based on Composite Risk Score

E. Explainable AI: Grad-CAM

To improve transparency and clinical trust, Gradient-weighted Class Activation Mapping (Grad-CAM) is used to highlight the CT image regions that have the most significant impact on the classification outcome. The Grad-CAM heatmap is calculated as:

$$L^c_{Grad\text{-}CAM} = \mathrm{ReLU}(\sum_k \alpha_k^c A^k) \quad (14)$$

where A^k is the activation map of the k-th convolutional feature map, and α_k^c is the weight of importance of the k-th feature map for target class c, calculated using global average pooling of the gradient:

$$\alpha_k^c = (1/Z) \sum_{i,j} \partial y^c / \partial A^k_{ij} \quad (15)$$

where Z is the total number of pixels in the feature map, and y^c is the classification score for class c. The ReLU function is used to ensure that only positive contributions are shown, resulting in a heatmap that is upscaled to the original image size for clinical interpretation..

F. Model Evaluation Metrics

The proposed model is assessed using the standard classification evaluation metrics. With TP (True Positives), TN (True Negatives), FP (False Positives), and FN (False Negatives):

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN) \quad (16)$$

$$\text{Precision} = TP / (TP + FP) \quad (17)$$

$$\text{Recall} = TP / (TP + FN) \quad (18)$$

$$\text{F1-Score} = 2 \cdot (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall}) \quad (19)$$

The AUC-ROC value is determined by integrating the Receiver Operating Characteristic curve, which calculates the model's ability to distinguish between malignant and benign images for all possible classification thresholds.

VI. Experimental Results

A comprehensive analysis of the proposed intelligent clinical framework was performed using the carefully compiled dataset of 1,000 pulmonary CT scan images (500 malignant and 500 benign).

A. Classification Performance Metrics

The performance metrics for classification are presented in Table I..

TABLE I — *Classification Performance Metrics*

Metric	Value (%)
Accuracy	95.2
Precision	94.8
Recall (Sensitivity)	95.6
F1-Score	95.2
AUC-ROC	96.1
Specificity	94.4

The proposed model was able to attain high classification performance for both malignancies and benignities on pulmonary CT scans. Specifically, 92% of high-risk patients were correctly classified by the risk stratification module, which clearly indicates the clinical value of the composite risk score.

B. Explainable AI Visualizations

The Grad-CAM heat maps were able to identify the relevant sites of anatomy that were used in the prediction generation. The heat maps made it easy for the clinicians to understand the logic behind the model, hence improving the model's transparency. Feature importance analysis further revealed that the most important features for all malignancies were nodule density variation and irregular boundary features

VII. Discussion

A. Classification Performance

The use of CNNs in feature extraction helped the algorithm to learn complex spatial and textural patterns of pulmonary nodules, achieving 95.2% accuracy and 96.1% AUC-ROC. The balanced performance of the algorithm on the metrics of precision, recall, and F1-score indicates its good generalization ability.

B. Clinical Risk Stratification

The tiered risk stratification scheme based on the composite risk score R has a certain practical use for clinicians. The timely identification of patients at high risk of the disease can prevent delays in diagnosis and lead to successful treatment outcomes.

C. Interpretability and Transparency

Applying Grad-CAM-based XAI methods for model explanation enhances the transparency of predictions. Pointing out key regions of interest and features related to pulmonary nodules helps in evidence-based decision-making and also aids in building clinician confidence in the automated model.

D. Comparative Analysis

The proposed framework is more accurate (3-5%) compared to the conventional CNN-based models for pulmonary cancer detection, and also has higher interpretability because of the XAI component. The inclusion of the composite risk score and CDSS recommendations makes this framework distinct from the existing literature.

E. Limitations

- The size of the dataset is small; larger datasets could potentially increase accuracy.
- The dataset only includes basic characteristics; genomic and longitudinal data could potentially increase accuracy.
- Validation of the model is necessary using real-world clinical data acquired in a variety of ways.

VIII. FUTURE WORK

- Incorporation of larger patient datasets from multiple institutions.
- Incorporation of genomic and molecular profiling data with clinical imaging data.
- Real-time integration with existing hospital clinical information systems.
- Use of federated learning approaches to facilitate collaboration between multiple institutions while maintaining patient privacy.
- Development of mobile applications for point-of-care delivery.

IX. Conclusion

This work sets up an intelligent clinical system for the prediction of pulmonary malignancy by employing AI technology to support clinical decisions. The system applies CNNs to feature extraction from CT images, a mathematically justified composite risk score for stratification, and XAI techniques (Grad-CAM) to produce predictions that are both accurate and interpretable. The

experimental outcome shows 95.2% accuracy and 96.1% AUC-ROC values, as well as correct identification of 92% of patients at high risk.

The proposed system advances current approaches that are only capable of detection by offering medical professionals a complete support system for clinical decisions that includes recommendations for therapy, prevention, and assistance with screening for organ donation. With its ability to make early and accurate detection, as well as offer clinically interpretable recommendations, the intelligent clinical system has the potential to improve the accuracy of diagnosis, help medical professionals offer better patient care, and achieve better patient outcomes.

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